

Jisang Lee

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Education

Hanbat National University (Advisor: Prof. Haneol Jang)

M.S. Student in Computer Engineering

Daejeon

Mar. 2025 - Present

Fourth semester in progress. **Cumulative GPA 4.5/4.5** through the first three semesters.

Hanbat National University (Advisor: Prof. Haneol Jang)

B.S. in Computer Engineering

Daejeon

Mar. 2019 - Feb. 2025

GPA 4.09/4.5 (Major: 4.28/4.5)

Publications

Conference

- [1] Eojin Jang, **Jisang Lee**, DongSu Kim, SeoYeon Oh, and Haneol Jang, “Development of Optical Character Recognition Model for Document Leak Investigation”, *2022 Korean Digital Forensic Society Winter Conference*, Dec 2022.
- [2] SeoYeon Oh, **Jisang Lee**, DongSu Kim, and Haneol Jang, “Lightweight Deep Convolutional Neural Networks for Fine-Grained Leaf Classification”, *The 33rd Joint Conference on Communications and Information*, Apr 2023.
- [3] **Jisang Lee**, DongSu Kim, SeoYeon Oh, and Haneol Jang, “Lightweight Fingerprint Forgery Detection Technique Utilizing Median Filter and Ensemble Model”, *2023 Korean Digital Forensic Society Summer Conference*, Jun 2023.
- [4] DongSu Kim, **Jisang Lee**, SeoYeon Oh, and Haneol Jang, “Effective Data Augmentations for Fingerprint Recognition”, *2023 Korean Digital Forensic Society Summer Conference*, Jun 2023.
- [5] HyeongWon Seo, **Jisang Lee**, DongSu Kim, SeoYeon Oh, and Haneol Jang, “Effective Detection of Generated Images Using Fast Fourier Transform and Discrete Cosine Transform”, *2023 Korean Digital Forensic Society Winter Conference*, Dec 2023.
- [6] **Jisang Lee**, HyeongWon Seo, Cheoneum Park, and Haneol Jang, “Crop-Paste Technique for Enhancing Object Detection Performance in Synthetic Aperture Radar Images”, *Korea Computer Congress 2024*, Jun 2024.
- [7] HyeongWon Seo, **Jisang Lee**, Cheoneum Park, and Haneol Jang, “Data Preprocessing and Augmentation Methods for SAR to Optical Image Translation”, *Korea Computer Congress 2024*, Jun 2024.
- [8] **Jisang Lee**[†], SeoYeon Oh[†], Cheoneum Park^{*}, and Haneol Jang^{*}, “Uncertainty-Guided Structural Routing and Consensus-Aware Re-Ranking for Open-Vocabulary Camouflaged Object Segmentation”, *Asian Conference on Computer Vision (ACCV)*, 2026. **To appear.** ([†]Equal contribution; ^{*}Corresponding authors.)

Journal

- [1] **Jisang Lee**, DongSu Kim, SeoYeon Oh, and Haneol Jang, “Lightweight Technique for Forged Fingerprint Detection Using Ensemble Models and Median Filters”, *Journal of Digital Forensics (KCI)*, 2023.
- [2] DongSu Kim, SeoYeon Oh, **Jisang Lee**, and Haneol Jang, “Effective Data Augmentations for Cross-Domain Fingerprint Recognition”, *Journal of Digital Forensics (KCI)*, 2024.
- [3] **Jisang Lee**, H. Kim, J. Koo, H. Choi, and D.Y. Jeong, “Parcel-Based Crop Type Classification in UAV Imagery with SAM for Smallholder Farms”, *Korean Journal of Remote Sensing (ESCI, Scopus)*, 2024.
- [4] HyeongWon Seo, DongSu Kim, SeoYeon Oh, **Jisang Lee**, and Haneol Jang, “Effective Detection of Generated Images Using Frequency Transform”, *Journal of KIISE (JOK)*, 2025.
- [5] **Jisang Lee**, HyeongWon Seo, WooJin Park, HyunCheol Choi, KyeongHo Kim, EunKyung Kim, and Haneol Jang, “Crop-Paste Data Augmentation Technique for Enhancing Object Detection Performance in Synthetic Aperture Radar Images”, *Journal of IKEEE (KCI)*, 2025.
- [6] J.H. Park, **Jisang Lee**, S.M. Pyo, and Haneol Jang, “Enhanced Copy-Paste Data Augmentation Using SAM for Precise Ship Detection”, *Journal of IKEEE (KCI)*, 2025.
- [7] HyeongWon Seo, **Jisang Lee**, WooJin Park, HyunCheol Choi, KyeongHo Kim, and Haneol Jang, “Effective Data Preprocessing and Augmentation Methods of SAR Images for Optical Image Translation”, *Journal of Broadcast Engineering (KCI)*, 2025.
- [8] **Jisang Lee**, HyeongWon Seo, EunKyung Kim^{*}, and Haneol Jang^{*}, “HFGF-DINOv3: High-Frequency Guided Fusion and Phase-Aware Loss for Camouflaged Object Detection”, *Pattern Recognition (SCIE)*, 2026. **Available online.** doi:10.1016/j.patcog.2026.115066. (^{*}Corresponding authors.)

Research Experience

Artificial Intelligence Multimodal Lab. (AIM Lab.), Hanbat National University

Daejeon, Republic of Korea

M.S. Researcher (Advisor: Prof. Haneol Jang)

Mar. 2025 - Present

- Research on detection and segmentation with vision foundation and vision-language models
- Focus on camouflaged objects, open-vocabulary recognition, and multispectral image understanding

SI Analytics Research Intern - Worked on UAV image segmentation and parcel-based crop classification - Improved a SAM-based parcel segmentation model; research published in KJRS 2024	Daejeon, Republic of Korea Jul. 2024 - Sep. 2024
AIM Lab., Hanbat National University Undergraduate Researcher (Advisor: Prof. Haneol Jang) Conducted research on computer vision and deep learning	Daejeon, Republic of Korea May 2022 - Feb. 2025

Research Projects

Deep Learning Research for X-ray Diagnosis of Neonatal Respiratory Distress Syndrome (RDS) Chungnam National University Hospital - Segmented X-ray images to isolate the lung regions - Enhanced performance through classification network training - Used Grad-CAM for analysis	Jan. 2023 - Jul. 2023
Research on SAR Image Analysis and Semantic Segmentation Capstone Design - Explored SAR image preprocessing techniques - Investigated despeckling and scaling techniques for SAR images - Devised generative models and data augmentation techniques for training data acquisition - Improved SAR segmentation performance through knowledge distillation	Jan. 2024 - Dec. 2024
Open-Vocabulary Zero-Shot Detection for Military Objects Collaborative Research with DataMaker - Conducted collaborative research on open-vocabulary zero-shot detection for military objects - Studied open-vocabulary recognition and detection strategies for unseen military targets - Built and analyzed experimental pipelines for robust zero-shot military object understanding - Investigated vision-language based approaches for military AI applications	Jan. 2025 - Oct. 2025

Awards and Honors

Competition Award	Special Prize (President’s Award), <i>2022 Creative Design Online Competition, Nanum Tech</i>	Nov 2022
Competition Award	Encouragement Prize (President of SW-Centered University Council Award), <i>SW-univ AI Competition 2022</i> (8th/77 teams)	Aug 2022
Best Paper Award	President’s Award, <i>Development of Optical Character Recognition Model for Document Leak Investigation</i>	Dec 2022
Competition Award	Prize, <i>Computer Engineering Portfolio Competition</i>	2022
Competition Result	1st/192 teams, <i>SW-Centered University Joint AI Competition 2022 – Preliminaries</i>	2022
Best Paper Award	Commissioner General Award(해양경찰청장상), <i>Lightweight Fingerprint Forgery Detection Technique Utilizing Median Filter and Ensemble Model</i>	Jun 2023
Competition Award	Special Prize (DEEPNOID Award; 17th/227 teams), <i>SW-univ AI Competition 2023</i> (위성 이미지 건물 영역 분할(Image Segmentation) AI모델 개발)	Aug 2023
Best Paper Award	President’s Award, <i>Effective Detection of Generated Images Using Fast Fourier Transform and Discrete Cosine Transform</i>	Dec 2023
Competition Award	Prize, <i>Computer Engineering Portfolio Competition</i>	2023
Competition Result	5th/542 teams, <i>Fake or Real: AI-Generated Image Detection Challenge, AI Connect</i>	2023
Competition Award	Grand Prize (1st/141 teams), <i>6th AI SPARK Challenge, Innopolis Foundation Chairman’s Award</i> (글로벌 산불 감지 챌린지: AI를 활용한 향상된 위성 이미지 분석)	Apr 2024
Competition Award	Gold Prize, <i>Capstone Design Exhibition (SkyPixel)</i>	2024
Competition Award	Prize, <i>Computer Engineering Portfolio Competition</i>	2024
Competition Award	Excellence Award (최우수상; 2nd/81 teams), <i>National Park Satellite Monitoring AI Challenge</i> Landslide area detection and risk analysis from satellite imagery	2026

Technical Skills

Programming	Python, Java, C, C++
Libraries	PyTorch, NumPy, Pandas, scikit-learn
Languages	Korean(Native), English(Beginner)
Certifications	Deep Learning Specialization, Coursera (five courses), 2023 LivDet 2023, Fingerprint Liveness Detection, Challenge 1 certificate

Teaching Experiences

Hanbat AI Intensive Course Teaching Assistant (Prof. Haneol Jang)	<i>Hanbat National University</i> Jan. 2023
Data Structure Class Teaching Assistant (Prof. Haneol Jang)	<i>Hanbat National University</i> Mar. 2023 - Jun. 2023
Artificial Intelligence and Applications Class Teaching Assistant (Prof. Haneol Jang)	<i>Hanbat National University</i> Mar. 2024 - Jun. 2024
Artificial Intelligence and Applications Class Teaching Assistant (Prof. Haneol Jang)	<i>Hanbat National University</i> Mar. 2025 - Jun. 2025
Computer Vision Teaching Assistant (Prof. Haneol Jang); 4th-year undergraduate course	<i>Hanbat National University</i> Fall 2025
Computer Vision Teaching Assistant (Prof. Haneol Jang); graduate course	<i>Hanbat National University</i> Spring 2026; Fall 2026 (ongoing)

Academic Service

IEEE Transactions on Multimedia (TMM) Reviewer	2026
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